

Assignment-Chapter 5 (Due 09/05/2026)

1. Multiple Choices. (For each question, please select the most appropriate answer.)

1.1. Which of the following complexes may undergo associative substitution reactions?

- (a) $\text{V}(\text{CO})_6$ (b) $\text{Re}(\text{CO})_4(\text{NO})$ (c) $\text{CpRh}(\text{CO})_2$
(d) two of them (e) all of them (f) none of them

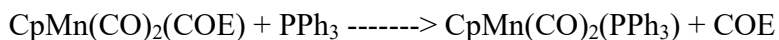
1.2. Which of the following reactions would you expect to be dissociative substitution reactions?

- (a) $\text{Pt}(\text{CO})_4 + \text{PMe}_3 \longrightarrow \text{Pt}(\text{CO})_3(\text{PMe}_3) + \text{CO}$
(b) $\text{HRh}(\text{CO})_4 + \text{PMe}_3 \longrightarrow \text{HRh}(\text{CO})_3(\text{PMe}_3) + \text{CO}$
(c) $\text{CH}_3\text{Re}(\text{CO})_5 + \text{PMe}_3 \longrightarrow \text{CH}_3\text{Re}(\text{CO})_4(\text{PMe}_3) + \text{CO}$
(d) $[\text{V}(\text{CO})_6]^- + \text{PMe}_3 \longrightarrow [\text{V}(\text{CO})_5(\text{PMe}_3)]^- + \text{CO}$
(e) two of them (f) three of them (g) all of them (h) none of them

1.3. Which of the following dissociative substitution reactions would you expect to have the lowest reaction rate?

- (a) $\text{Ru}(\text{CO})_4(\text{PEt}_3) + \text{CO} \longrightarrow \text{Ru}(\text{CO})_5 + \text{PEt}_3$
(b) $\text{Ru}(\text{CO})_4(\text{OEt}_2) + \text{CO} \longrightarrow \text{Ru}(\text{CO})_5 + \text{OEt}_2$
(c) $\text{W}(\text{CO})_5(\text{PEt}_3) + \text{CO} \longrightarrow \text{W}(\text{CO})_6 + \text{PEt}_3$
(d) $\text{W}(\text{CO})_5(\text{OEt}_2) + \text{CO} \longrightarrow \text{W}(\text{CO})_6 + \text{OEt}_2$

2. Propose a reaction mechanism for the following reaction



Your mechanism must be consistent with the following observations:

- (a) using excess PPh_3 , the rate is first order in Rh complex.
(b) $\Delta S^\ddagger = -17.9 \text{ eu}$

